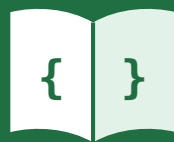

Module 02 — Configuration and Auto-Configuration

Exercises

Spring Boot Edu



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How to Work

1. The starter code is in `modules/02-configuration/exercise/`. Find the TODO comments (Java files, `application.yaml` and `META-INF/spring/...imports`).
2. Every exercise already has tests, and they stay **red** until you solve it.
3. Run the tests:

```
./mvnw -Pexercises -pl modules/02-configuration/exercise -am test
```

4. The exercise is done when all tests are green.
5. If you get stuck, read the hints first, then the solution in `modules/02-configuration/solution/`.

★ Tip

To run the tests of a single exercise: `-Dtest=Exercise1Test -Dsurefire.failIfNoSpecifiedTests=false`

Exercise 1 — Validated Opening Hours (Easy)

Goal: write a typed and validated `@ConfigurationProperties` record.

The bookstore's opening hours are defined under `bookstore.opening-hours.*`. An invalid setting (e.g. closing before opening) must stop the application from starting.

Tasks (`exercise1.OpeningHoursProperties`):

- TODO 1a — mark the record so that it is validated at startup.
- TODO 1b — make `opens` and `closes` required.
- TODO 1c — `maxReservations` must be between 1 and 10, and 3 when not set.
- TODO 1d — `closedDays` must be an empty list when not set.
- TODO 1e — validation must fail when the closing time is not after the opening time.
- TODO 1f — `isOpen(day, time)`: true on days that are not closed, from opening (inclusive) to closing (exclusive).

Hints:

- Lesson section 3.2: `@Validated`, `@NotNull`, `@Min/@Max`, `@DefaultValue`.
- For a cross-field rule, put `@AssertTrue(message = "...")` on a boolean method named `isXxx()`. Values may be null, so check that first.

Acceptance criteria: all 5 tests in `Exercise1Test` pass.

Estimated time: 30 minutes

Exercise 2 — Pricing per Profile (Medium)

Goal: produce different prices with the same code, using profile files and a profile group.

PricingProperties (bookstore.pricing.*) is provided. You will write the values into the YAML files.

Tasks (exercise/src/main/resources/):

- TODO 2a — application.yaml: default shipping fee 29.90, discount 0.
- TODO 2b — application-campaign.yaml: shipping 0.00, discount 20. application-express.yaml: shipping 49.90.
- TODO 2c — a profile group named black-friday that activates campaign first, then express.

Hints:

- Lesson section 3.4: spring.profiles.group.<name>: profile1, profile2.
- Within a group, the profile that comes **later** wins for the same key. That is why on black-friday the discount comes from campaign and the shipping fee from express.
- Write money values in quotes: "29.90".

Acceptance criteria: all 4 tests in Exercise2Test pass.

Estimated time: 30 minutes

Exercise 3 — Your Own Auto-Configuration (Hard)

Goal: write an auto-configuration that is guarded by conditions, backs off when the user has a bean, and is found through the imports file.

ExchangeRateService, FixedExchangeRateService and ExchangeRateProperties (bookstore.exchange.*) are provided. What is missing is the auto-configuration that brings them together.

Tasks (exercise3.ExchangeRateAutoConfiguration and the imports file):

- TODO 3a — make the class an auto-configuration.
- TODO 3b — it must apply only when bookstore.exchange.enabled is true or not set at all.
- TODO 3c — bind ExchangeRateProperties.
- TODO 3d — create a FixedExchangeRateService bean from the configured rates. Do not create it when the application has its own ExchangeRateService.
- TODO 3e — register the class in META-INF/spring/org.springframework.boot.autoconfigure.AutoCo

Hints:

- GreetingAutoConfiguration in lesson section 3.6 is a one-to-one example.

- `@ConditionalOnBooleanProperty(name = ..., matchIfMissing = true), @EnableConfigurationProperties, @ConditionalOnMissingBean`.
- Write the **fully qualified** class name (with the package) into the imports file.

Acceptance criteria: all 5 tests in `Exercise3Test` pass.

Estimated time: 45 minutes

Bonus Challenge (Optional)

Split Exercise 3 into two separate Maven modules (`...-autoconfigure` and `...-spring-boot-starter`) as in lesson section 3.6, and use the starter from the lesson application. Check that your IDE auto-completes `bookstore.exchange.rates.USD`.